

Assignment 2 : Installation and Maintenance of Hardware and Its Components

Section 1: Multiple Choice

1. Which of the following precautions should be taken before working on computer hardware?

Answer: b) Wear an anti-static wrist strap to prevent damage from electrostatic discharge.

2. What is the purpose of thermal paste during CPU installation?

Answer: c) To improve thermal conductivity between the CPU and the heat sink.

3. Which tool is used to measure the output voltage of a power supply unit (PSU)?

Answer: a) Multimeter

4. Which component is responsible for storing BIOS settings, such as date and time, even when the computer is powered off?

Answer: a) CMOS battery

Section 2: True or false

5. True or False: When installing a new hard drive, it is essential to format it before use.

Answer: True

6. True or False: A POST (Power-On Self-Test) error indicates a problem with the CPU.

Answer: False – It can indicate issues with several components, not only the CPU.

7. True or False: It is safe to remove a USB flash drive from a computer without ejecting it first.

Answer: False – Ejecting prevents data corruption.

Section 3: Short Answer

8. Describe the steps involved in installing a new graphics card in a desktop computer.

Answer:

- Power off and unplug the computer.
- Open the cabinet and locate the PCIe x16 slot.
- Remove the old GPU (if any) and unscrew the bracket.
- Insert the new graphics card firmly into the PCIe slot.
- Screw it into the bracket for support.
- Connect PCIe power connectors if required.
- Close the cabinet, power on, and install drivers.

9. What is RAID, and what are some common RAID configurations?

Answer:

RAID (Redundant Array of Independent Disks) is a storage technology that combines multiple drives for performance or redundancy.

Common RAID types:

- RAID 0 – Striping (performance)
- RAID 1 – Mirroring (redundancy)

- RAID 5 – Striping + parity (balanced)
- RAID 10 – Mirroring + striping (high speed + redundancy)

Section 4: Practical Application

10. Demonstrate how to replace a CPU fan in a desktop computer.

Answer:

- Turn off and unplug the PC.
- Open the case and locate the CPU cooler.
- Disconnect the fan's power cable from the motherboard.
- Unscrew or unclip the existing CPU fan assembly.
- Clean old thermal paste from the CPU.
- Apply new thermal paste.
- Install the new CPU fan and secure it.
- Connect its power cable and close the case.

Section 5: Essay

11. Discuss the importance of regular maintenance for computer hardware and provide examples of maintenance tasks.

Answer:

Regular maintenance improves performance, prevents overheating, extends hardware life, and reduces failures.

Importance of Regular Maintenance

1. Improves Performance:

Dust buildup and outdated drivers can slow down the system.

Regular cleaning and updates help maintain smooth performance.

2. Prevents Overheating:

Accumulated dust blocks airflow in cooling fans and vents, causing the computer to heat up. Regular cleaning ensures proper cooling.

3. Increases Hardware Lifespan:

Components like CPU, GPU, RAM, and hard drives last longer when they operate under normal temperatures and clean conditions.

4. Reduces Risk of Data Loss:

Maintaining storage devices and regularly checking for errors helps prevent corruption and accidental data loss.

5. Enhances System Stability:

Updated BIOS, firmware, and drivers ensure compatibility with new software and reduce crashes or malfunctions.

6. Cost-Effective:

Preventive maintenance avoids expensive repairs by identifying issues early (e.g., failing fans, loose cables, weak PSU).